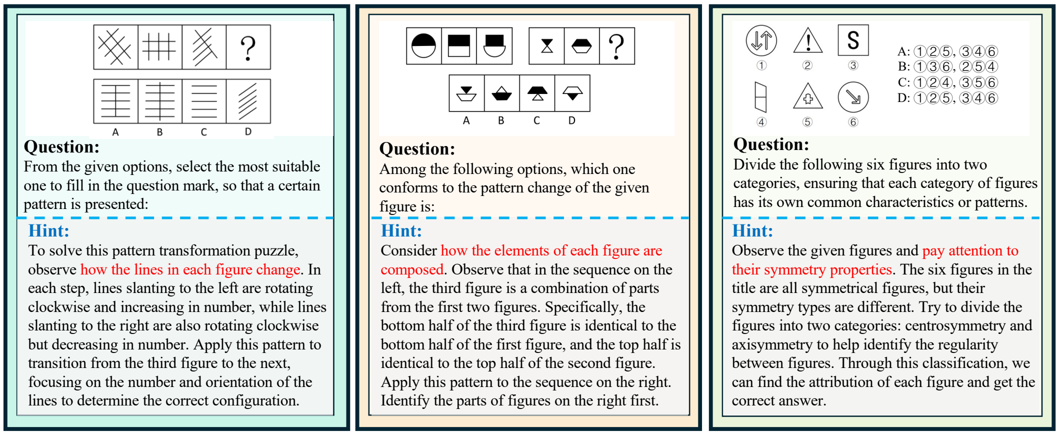


Figure 7: **Hint prompts visualization.** Hint prompts examples, which supply solution guidance for MLLMs, are shown in the image, with solution-critical elements highlighted in red.

tical datasets, RL-enhanced models demonstrate substantially larger performance gains, underscoring the promise of targeted RL methods for advancing multimodal visual reasoning.

4.3 Fine-grained Comparison

We systematically analyze model capabilities by examining error distributions across reasoning categories for different models. Figure 8 presents the error rates of LLMs, MLLMs, and human participants over six distinct reasoning categories.

Figure 8a reveals that LLMs struggle most with *Spatial Reasoning* questions, indicating that text-only descriptions are insufficient to infer three-dimensional structures or spatial transformations. In contrast, their performance on *Quantitative Reasoning* tasks is comparatively stronger, suggesting that quantitative relationships are more readily conveyed through language.

As shown in Figure 8b, *Stylistic Reasoning* presents the greatest difficulty for MLLMs, with error rates exceeding 75%—worse than random guessing (25% accuracy). This result underscores a fundamental limitation of current MLLM architectures in capturing subtle visual cues such as overlays, contours, and shape variations.

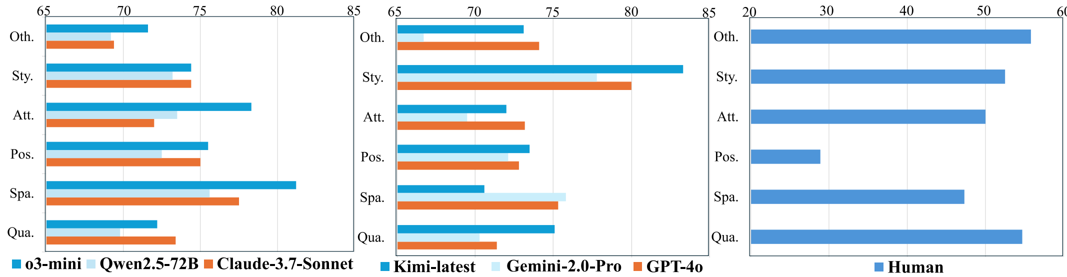
Figure 8c reveals that human error patterns form a distinct cluster, separate from both LLMs and MLLMs. Human participants maintain error rates below 30% on *Positional Reasoning* tasks, reflecting robust position-based visual inference. By contrast, both model classes struggle with positional reasoning, highlighting a fundamental divergence in visual–cognitive processes between humans and MLLMs.

4.4 Qualitative Analysis

LLM Failures. As shown in Figure 6(h), text-only LLMs that rely on externally generated image captions often omit critical visual details required for multi-step logical deduction—such as the counts, shapes, and progression patterns of the black and white dots in Figure 6(a). Consequently, their reasoning diverges from the correct solution and frequently yields hallucinations or irrelevant responses.

MLLM Failures. Figure 6 also presents cases in which MLLMs correctly describe static visual content yet fail to infer the evolving relationships among shapes, instead resorting to superficial cues like object counts. While these models can recognize individual shapes and tally items, they struggle to reason over inter-element relations, which limits their ability to solve complex visual-logic problems.

RL-Based Improvements. As illustrated in Figure 6(g), reinforcement learning (RL) encourages deeper, stepwise logical reasoning. The RL-enhanced model successfully captures state transitions (e.g., the movements of chess pieces in Figure 6(a)) and accurately predicts subsequent configurations. Moreover, it learns to iteratively revise intermediate hypotheses—akin to trial-and-error—until a



(a) LLMs' error distribution. (b) MLLMs' error distribution. (c) Humans' error distribution.

Figure 8: **Error distribution analysis.** The figure demonstrates distinct error type allocations across Humans, LLMs and MLLMs, revealing differences among their cognition patterns.

coherent deduction emerges (see additional examples in the Appendix). These findings highlight the potential of RL methods to bolster performance on visual reasoning tasks.

5 Conclusion

In this paper, we present VisuLogic, a novel benchmark designed to evaluate the visual reasoning capabilities of Multi-modal Large Language Models (MLLMs). The benchmark consists of 1,000 vision-centric reasoning tasks distributed across six distinct categories. We conduct comprehensive evaluation of several advanced LLMs and MLLMs on this benchmark and provide an in-depth analysis of their performance. Our findings reveal that even the most advanced models fall short of human performance, highlighting substantial opportunities for advancement in visual logical reasoning. Through further experiments, we find that reinforcement learning (RL) is a promising approach for enhancing the vision reasoning capabilities of MLLMs. To promote further research and innovation, we open-source the evaluation code, training scripts, and datasets associated with this work.

References

[1] J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F. L. Aleman, D. Almeida, J. Altenschmidt, S. Altman, S. Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.

[2] W. U. Ahmad, S. Narenthiran, S. Majumdar, A. Ficek, S. Jain, J. Huang, V. Noroozi, and B. Ginsburg. Opencodereasoning: Advancing data distillation for competitive coding. *arXiv preprint arXiv:2504.01943*, 2025.

[3] A. Ahmadian, C. Cremer, M. Gallé, M. Fadaee, J. Kreutzer, O. Pietquin, A. Üstün, and S. Hooker. Back to basics: Revisiting reinforce style optimization for learning from human feedback in llms, 2024.

[4] S. N. Akter, S. Lee, Y. Chang, Y. Bisk, and E. Nyberg. Visreas: Complex visual reasoning with unanswerable questions, 2024.

[5] J.-B. Alayrac, J. Donahue, P. Luc, A. Miech, I. Barr, Y. Hasson, K. Lenc, A. Mensch, K. Millican, M. Reynolds, et al. Flamingo: a visual language model for few-shot learning. *Advances in neural information processing systems*, 35:23716–23736, 2022.

[6] H. Bai, Y. Zhou, L. E. Li, S. Levine, and A. Kumar. Digi-q: Learning q-value functions for training device-control agents. *arXiv preprint arXiv:2502.15760*, 2025.

[7] J. Bai, S. Bai, S. Yang, S. Wang, S. Tan, P. Wang, J. Lin, C. Zhou, and J. Zhou. Qwen-vl: A versatile vision-language model for understanding, localization, text reading, and beyond. *arXiv preprint arXiv:2308.12966*, 2023.

[8] S. Bai, K. Chen, X. Liu, J. Wang, W. Ge, S. Song, K. Dang, P. Wang, S. Wang, J. Tang, H. Zhong, Y. Zhu, M. Yang, Z. Li, J. Wan, P. Wang, W. Ding, Z. Fu, Y. Xu, J. Ye, X. Zhang, T. Xie, Z. Cheng, H. Zhang, Z. Yang, H. Xu, and J. Lin. Qwen2.5-vl technical report. *arXiv preprint arXiv:2502.13923*, 2025.

[9] G. Chen, M. Wang, Y. Yang, K. Yu, L. Yuan, and Y. Yue. Pointgpt: Auto-regressively generative pre-training from point clouds. *Advances in Neural Information Processing Systems*, 36:29667–29679, 2023.

[10] L. Chen, J. Li, X. Dong, P. Zhang, C. He, J. Wang, F. Zhao, and D. Lin. Sharegpt4v: Improving large multi-modal models with better captions. *arXiv preprint arXiv:2311.12793*, 2023.

[11] L. Chen, L. Li, H. Zhao, Y. Song, and Vinci. R1-v: Reinforcing super generalization ability in vision-language models with less than $3. `https://github.com/Deep-Agent/R1-V`, 2025. Accessed: 2025-02-02.

[12] S. Chen, H. Li, Q. Wang, Z. Zhao, M. Sun, X. Zhu, and J. Liu. Vast: A vision-audio-subtitle-text omni-modality foundation model and dataset. *Advances in Neural Information Processing Systems*, 36:72842–72866, 2023.

[13] X. Chen, H. Fang, T.-Y. Lin, R. Vedantam, S. Gupta, P. Dollar, and C. L. Zitnick. Microsoft coco captions: Data collection and evaluation server, 2015.

[14] Z. Chen, W. Wang, Y. Cao, Y. Liu, Z. Gao, E. Cui, J. Zhu, S. Ye, H. Tian, Z. Liu, et al. Expanding performance boundaries of open-source multimodal models with model, data, and test-time scaling. *arXiv preprint arXiv:2412.05271*, 2024.

[15] Z. Chen, W. Wang, H. Tian, S. Ye, Z. Gao, E. Cui, W. Tong, K. Hu, J. Luo, Z. Ma, et al. How far are we to gpt-4v? closing the gap to commercial multimodal models with open-source suites. *arXiv preprint arXiv:2404.16821*, 2024.

[16] Z. Chen, J. Wu, W. Wang, W. Su, G. Chen, S. Xing, M. Zhong, Q. Zhang, X. Zhu, L. Lu, et al. Internvl: Scaling up vision foundation models and aligning for generic visual-linguistic tasks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 24185–24198, 2024.

[17] S. Cheng, Z. Guo, J. Wu, K. Fang, P. Li, H. Liu, and Y. Liu. Egothink: Evaluating first-person perspective thinking capability of vision-language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14291–14302, 2024.

[18] DeepSeek-AI. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning, 2025.